Code-Layout, Readability and Reusability

Date	27 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Python code follows PEP-8 indentation standards.
2	Proper File Structure	Files and folders are logically organized	Yes	Clear separation of templates/, static/, models/, and app.py.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as life_expectancy, schooling, gni, and predicted_hdi improve readability.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as index(), predict(), classify_hdi(), and get_tier_info() are clearly named.
5	Code Comments	Inline and block comments explain logic	Yes	The application includes meaningful comments and function docstrings to improve readability and maintenance.
6	Modular Design	Code is split into reusable functions/modules	Yes	The trained model, scaler, and label encoder are saved separately and loaded as reusable modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Common logic is reused to reduce code duplication.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling prevent application crashes..

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Trained ML Model (hdi_model.pkl)	Python / Scikit-learn	Used for every prediction request	High
2	Flask Application	Python / Flask	Handles prediction requests	High
3	Prediction Route (/predict)	Python / Flask	Processes user input and generates HDI predictions.	High
4	Frontend Templates (index.html & result.html)	HTML ,CSS, JavaScript	Used for collecting inputs and displaying prediction results	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	**5**	Well-organized project with separate frontend, backend, and model files.
Readability	**5**	Clear variable names, readable functions, and consistent formatting.
Reusability	**5**	Model, preprocessing objects, and Flask application components can be reused in future applications.
Documentation / Comments	**4**	Important sections are documented with comments for better understanding.
Overall Score	**4.6/5**	Clean, modular, and maintainable machine learning application.